NARESH B A

Backend Engineer | AI Systems | Distributed Systems

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medium.com/@phoenixarjun007 | Chennai, India (Open to Remote)

Backend and AI systems engineer with 2+ years building production distributed systems and intelligent pipelines. Designed large-scale event-driven architectures (Kafka, Saga, CDC) and multi-agent LLM systems with real observability. Published open-source tooling on npm, PyPI, and Cargo. Active technical writer at Medium — 15+ articles on distributed systems, AI engineering, and RAG design.

Skills

Languages Python, Java, TypeScript, SQL
AI / LLM LangChain, LangGraph, CrewAI, MCP, A2A Protocol, RAG Pipelines, Multi-Agent Systems, PyTorch
Backend Spring Boot, FastAPI, Flask, REST, gRPC, JWT, Hexagonal Architecture
Distributed Apache Kafka, Debezium CDC, Saga Pattern, Outbox Pattern, CQRS, Event-Driven Architecture
Databases PostgreSQL, MongoDB, Elasticsearch, Redis, MySQL, pgvector, Qdrant
Infra / Cloud Docker, Kubernetes, GCP, Helm, GitHub Actions
Observability OpenTelemetry, Prometheus, Grafana, Jaeger, ELK Stack, LangFuse

Experience

Software Development Engineer July 2025 – Present
Prodapt Solutions Chennai, India

- Architected a **TSDS migration pipeline** for Elasticsearch re-indexing **800M+ documents per backing index** from standard to time-series format, achieving **~30% storage reduction** (900 GB → ~550 GB per day-index); implemented snapshot-first strategy to GCS for zero-risk historical migration.
- Engineered a **distributed alert processing system** using Confluent Kafka (24 partitions, 24 pods, 200 threads/pod) processing **10M records in under 6 minutes**; reduced payload volume from 10M to 6–7M through upstream threshold caching.
- Built a **recursive MIB dependency resolution engine** for SNMP MIB processing auto-detecting failed conversions, persisting raw files to GCS, and re-triggering conversion upon dependency arrival; handles **6 levels of dependency chaining** in production, eliminating the need to upload full dependency sets manually.
- Designed an **Elasticsearch ILM hot/warm/cold tiering pipeline** for 80TB/day telemetry data targeting **60%+ storage reduction** via automated re-indexing and lifecycle optimisation. Recognised with **Rising Rookie of the Month Award**.

Projects

PromNova — AI-Powered Prompt Engineering Platform | *Python · FastAPI · LangChain · LangGraph · Next.js · Gemini · Groq*

- Built a **multi-agent prompt generation engine** using a fan-out strategy across **25+ prompting types and 18+ frameworks** (zero-shot, few-shot, chain-of-thought, Tree-of-Thought, CO-STAR, RISEN); each selected type triggers a **parallel LLM agent call**, with outputs aggregated and scored by an evaluator agent against a quality threshold a refinement agent iterates up to 3 loops before final delivery.
- Implemented **LLM-as-Judge evaluation** combined with a T-RAG framework for goal-alignment scoring across multiple quality dimensions; built a structured JSON prompt template generator accepting project intent and user-defined variables for LangGraph-based agent projects.

Architecture As Memory (AAM) — Open-Source Architectural Cognition | *TypeScript · Node.js · React Flow · YAML · Zod · NPX CLI*

- Designed and published an **open-source framework** (npm, PyPI, Cargo) that treats software architecture as **persistent AI-readable memory** solving cognitive drift in AI-assisted development by maintaining a capability-based YAML model of systems, domains, components, and relationships that evolves alongside the codebase.
- Built an **interactive React Flow visualiser** with hierarchical expansion, architectural smell detection, deterministic topology hashing, snapshot history, and offline export; integrated with **Claude Code, Gemini CLI, Cursor, and Codex** via provider-aware instruction injection and slash commands zero-config NPX install.

Publications & Writing

Active author at medium.com/@phoenixarjun007 — 15+ articles on distributed systems, AI agent design, and LLM engineering.

- *Kafka Guarantees Delivery, Not Uniqueness: How to Build Idempotent Systems*
- *Why Most RAG Systems Hallucinate — And How a Hybrid Pipeline Fixes It*
- *Beyond Prompt Engineering: The Layers of Modern AI Engineering*

Education

B.Tech — Information Technology Graduated May 2025
Sri Sai Ram Institute of Technology | Anna University CGPA: 8.44

Certifications

Oracle Cloud Infrastructure AI Foundations · Oracle Certified Data Scientist · Apache Kafka for Beginners v3 — Udemy (Stephane Maarek) · MLOps Specialisation — DeepLearning.AI